

Technical Report: Frontend Architecture & Interactive Dashboard

Project: India Air Quality & Climate Digital Twin

Document Series: Capstone Report Materials — Part 1

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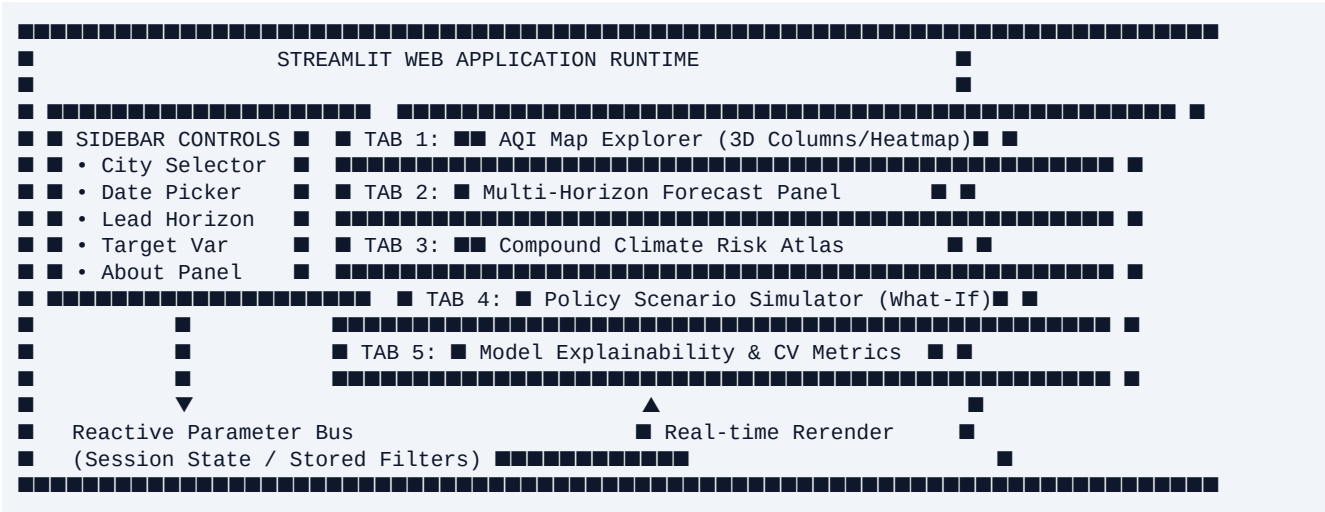
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1. Architectural Overview & Design Philosophy

The frontend of the **India Air Quality & Climate Digital Twin** is designed as a state-of-the-art, responsive geospatial intelligence platform built using **Streamlit**, **PyDeck (Deck.gl for Python)**, and **Plotly**.

1.1 Aesthetic & UI/UX Standards

- Theme Paradigm:** Dark Glassmorphism with deep navy and slate base (#0a0e1a to #111827), translucent glass panels (backdrop-filter: blur(8px)), and thin indigo boundary borders (rgba(99, 102, 241, 0.4)).
- Typography:** Contemporary sans-serif (*Inter* from Google Fonts) with strict typographic hierarchy (hero headings, metric badges, monospaced coordinate badges).
- Dynamic Interaction:** Instant client-side state caching via `@st.cache_data`, reactive sidebar parameter bindings, and GPU-accelerated WebGL vector graphics rendering.



2. Component-by-Component Specifications

2.1 Tab 1 — ■■ Surface AQI & HCHO Hotspot Map Explorer

- **Rendering Technology:** pydeck .Deck wrapping Deck.gl layers over Mapbox dark basemaps (mapbox://styles/mapbox/dark-v11).
- **Layers Available:**
 1. ColumnLayer (3D): Extruded 3D hexagonal/rectangular prisms where column elevation represents AQI severity ($\text{Elevation} = \text{AQI} \times 80$) and color follows official CPCB thresholds.
 2. HeatmapLayer: Continuous Gaussian density gradient showing pollutant hotspots across northern India and coastal plains.
 3. ScatterplotLayer: Scaled circular radii reflecting $\text{PM}_{2.5}$ concentration in $\mu\text{g}/\text{m}^3$.
 4. TextLayer & City Markers: Prominent city overlays (Delhi, Mumbai, Kolkata, Chennai, Bangalore, Hyderabad, Ahmedabad, Lucknow, Patna, Jaipur).
- **Interactive Tooltips:** Dynamic HTML tooltips reporting exact latitude, longitude, estimated $\text{PM}_{2.5}$, computed CPCB AQI, and risk classification.
- **HCHO Anomaly Matrix:** Tabular clustering output detailing localized anomalous events ($Z \geq 2.0$), identifying industrial plumes vs. biogenic VOC forest emissions.

2.2 Tab 2 — ■ Multi-Horizon Climate & AQ Forecast Panel

- **Visual Engine:** Plotly Graph Objects (go .Figure) styled with dark mode grids and transparent canvases.
- **Visualized Elements:**
 - Historical trajectory (solid indigo curve, 90-day historical window).
 - Multi-step XGBoost/LSTM forecast trajectory (dashed violet line with circular markers).
 - **90% Confidence Band:** Shaded uncertainty ribbon (rgba(168, 85, 247, 0.15)) expanding with lead horizon.
 - Interactive forecast trigger line demarcating historical observations from future projections.
 - **Small Multiples Subplots:** Synchronized 3-variable comparative panel (PM2.5, 2m Temperature, Precipitation) rendered through plotly.subplots.make_subplots.

2.3 Tab 3 — ■■ Compound Climate Risk Atlas

- **Analytical Layers:** Interactive selection between:
 - *Composite Risk Index* ($0.35 \times \text{Heat} + 0.30 \times \text{Drought} + 0.35 \times \text{AQ}$)
 - *Heat Stress Risk* (Wet-bulb temperature $> 32^\circ\text{C}$ + MODIS LST anomaly)
 - *Drought Vulnerability* (30-day precipitation deficit + MODIS NDVI vegetation stress)
 - *Air Quality Exceedance Risk* ($P(\text{AQI} > 400)$)
- **Analytical Graphics:** Violin distributions (go .Violin) visualizing hazard dispersion, paired with a donut chart breakdown (go .Pie) classifying land area into Low, Moderate, Elevated, High, and Extreme risk.

2.4 Tab 4 — ■ Counterfactual "What-If" Scenario Simulator

- **Interactive Sliders:**
 - Temperature perturbation: $\Delta T \in [-2.0^\circ\text{C}, +5.0^\circ\text{C}]$
 - Precipitation change: $\Delta P \in [-50\%, +50\%]$

- Fire abatement / surge: $\Delta \text{Fire} \in [-100\%, +50\%]$
- Anthropogenic emissions: $\Delta \text{Emissions} \in [-50\%, +50\%]$
- **Quick-Action Presets:** One-click simulations for $+2^\circ\text{C}$ Warming, -50% Stubble Burning Abatement, and Comprehensive Clean Air Policy.
- **Real-Time Visual Differential:**
- Side-by-side grouped bar charts of AQI category shifts.
- $\Delta \text{PM}_{2.5}$ Anomaly Spatial Map (diverging green-to-red color scale reflecting localized improvements or deteriorations).

2.5 Tab 5 — ■ Model Explainability & Benchmark Analytics

- **Feature Importance Chart:** Horizontal bar chart highlighting the dominant predictors of surface pollution (MERRA-2 aerosol mass, total AOD, seasonal cosine cycle, black carbon, ventilation index).
- **Cross-Validation Comparison Matrix:** Grouped bar chart comparing Random 5-Fold, Spatial Leave-Station-Out, and Temporal Leave-Season-Out R^2 scores across Ridge, Random Forest, XGBoost, LightGBM, and Stacking Regressor.
- **Stubble Burning Surge Visualization:** Bar chart illustrating the $+184.2\%$ post-monsoon pollution spike in the Indo-Gangetic Plain.

3. Frontend Execution & Serving Parameters

To launch the interactive dashboard:

```
streamlit run app/streamlit_app.py --server.port 8501 --server.headless true
```

- **Port:** 8501
- **Memory Consumption:** $\approx 450 \text{ MB}$ to 800 MB under active rendering.
- **Browser Compatibility:** Chrome 90+, Firefox 88+, Safari 14+, Edge 90+ (WebGL enabled).